

Processando config 1, seed 1, teste: BlockAlphabit
xx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xx

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 9.37e-3
p-value of test : 0.9953

CPU time used : 00:00:00.02

Generator state:
16777216 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 8.40
p-value of test : 0.40

CPU time used : 00:00:00.01

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 118.40
p-value of test : 0.72

CPU time used : 00:00:00.02

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33163.89
p-value of test : 0.06

CPU time used : 00:00:00.02

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 8887.69
p-value of test : eps *****

CPU time used : 00:00:00.01

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305	
Chi-square statistic	:	3153.30	
p-value of test	:	eps	*****

CPU time used : 00:00:00.03

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	-1.47	
p-value of test	:	0.93	

CPU time used : 00:00:00.01

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32	
ChiSquare statistic	:	765.07	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	31	
ChiSquare statistic	:	1690.45	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	32	
ChiSquare statistic	:	435.49	
p-value of test	:	eps	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	25	
ChiSquare statistic	:	60.03	
p-value of test	:	1.0e-4	*****

Test on the values of the Statistic C

Number of degrees of freedom	:	15	
ChiSquare statistic	:	34.59	
p-value of test	:	2.8e-3	

CPU time used : 00:00:00.06

Generator state:

134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	593.95	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	566.15	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	298.21	
p-value of test	:	2.1e-10	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	65.49	
p-value of test	:	0.21	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	16.22	
p-value of test	:	0.9907	

CPU time used : 00:00:00.06

Generator state:

150994944 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.31
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):

(eps1 means a value < 1.0e-15):

	Test	p-value
5	HammingIndep, L = 16	eps
6	HammingIndep, L = 32	eps
8	RandomWalk1 H (L = 64)	eps
8	RandomWalk1 M (L = 64)	eps
8	RandomWalk1 J (L = 64)	eps
8	RandomWalk1 R (L = 64)	1.0e-4
9	RandomWalk1 H (L = 320)	eps
9	RandomWalk1 M (L = 320)	eps
9	RandomWalk1 J (L = 320)	2.1e-10

All other tests were passed

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XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 3.27
p-value of test                   : 0.20
```

```
-----
CPU time used                     : 00:00:00.03
```

Generator state:
167772160 bits have been read.

```
*****
```

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 7.13
p-value of test : 0.52

CPU time used : 00:00:00.00

Generator state:

184549376 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128

Value of the statistic : 86.27
p-value of test : 0.9982

CPU time used : 00:00:00.01

Generator state:
201326592 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33162.85
p-value of test : 0.06

CPU time used : 00:00:00.03

Generator state:
218103808 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 177  
Chi-square statistic              : 162.56  
p-value of test                   : 0.77  
-----
```

```
-----  
CPU time used                     : 00:00:00.01  
-----
```

Generator state:

234881024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

```
-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0  
-----
```

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 305  
Chi-square statistic              : 309.67  
p-value of test                   : 0.41  
-----
```

```
-----  
CPU time used                     : 00:00:00.00  
-----
```

Generator state:

251658240 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingCorr test:

```
-----  
N = 1, n = 524288, r = 0, s = 32, L = 32  
-----
```

```
-----  
Normal statistic                  : -1.94  
p-value of test                   : 0.97  
-----
```

CPU time used : 00:00:00.01

Generator state:
268435456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	41.37
p-value of test	:	0.12

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	36.08
p-value of test	:	0.24

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	49.40
p-value of test	:	0.03

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	25.36
p-value of test	:	0.44

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	21.47
p-value of test	:	0.12

CPU time used : 00:00:00.07

Generator state:
285212672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	807.97	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	365.25	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	263.99	
p-value of test	:	4.3e-7	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	59.99	
p-value of test	:	0.37	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	18.48	
p-value of test	:	0.97	

CPU time used : 00:00:00.06

Generator state:
301989888 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.28
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	4.3e-7

All other tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	0.29
p-value of test	:	0.86

CPU time used : 00:00:00.02

Generator state:
318767104 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 10.63
p-value of test : 0.22

CPU time used : 00:00:00.00

Generator state:
335544320 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256

Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 133.20
p-value of test : 0.36

CPU time used : 00:00:00.01

Generator state:
352321536 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32652.07
p-value of test : 0.67

CPU time used : 00:00:00.02

Generator state:
369098752 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 183.49
p-value of test : 0.35

CPU time used : 00:00:00.01

Generator state:

385875968 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 6344.63
p-value of test : eps *****

CPU time used : 00:00:00.00

Generator state:

402653184 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.21
p-value of test : 0.42

CPU time used : 00:00:00.01

Generator state:

419430400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32
ChiSquare statistic : 2685.29
p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 31
ChiSquare statistic : 812.10
p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom : 32
ChiSquare statistic : 130.81
p-value of test : 6.7e-14 *****

Test on the values of the Statistic R

Number of degrees of freedom : 25
ChiSquare statistic : 37.66
p-value of test : 0.05

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	15.51
p-value of test	:	0.42

CPU time used : 00:00:00.08

Generator state:
436207616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	959.66	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	438.80	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	308.75	
p-value of test	:	1.6e-11	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	48.79
p-value of test	:	0.77

Test on the values of the Statistic C

Number of degrees of freedom : 32
ChiSquare statistic : 40.93
p-value of test : 0.13

CPU time used : 00:00:00.05

Generator state:
452984576 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.26
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

	Test	p-value
6	HammingIndep, L = 32	eps
8	RandomWalk1 H (L = 64)	eps
8	RandomWalk1 M (L = 64)	eps
8	RandomWalk1 J (L = 64)	6.7e-14
9	RandomWalk1 H (L = 320)	eps
9	RandomWalk1 M (L = 320)	eps
9	RandomWalk1 J (L = 320)	1.6e-11

All other tests were passed

XX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216

Number of cells = $2^L =$ 4
Expected number per cell = 4194304
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

$\Delta = 1$, $\mu = 2.9802323e-08$, $\sigma = 1$

Test Results for $\Delta = 1.0000$

Number of degrees of freedom	:	2
Value of the statistic	:	3.54
p-value of test	:	0.17

CPU time used : 00:00:00.01

Generator state:
469761792 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

 $N = 1$, $n = 16777216$, $r = 0$, $s = 32$, $L = 4$,
Sparse = FALSE

Number of bits = $n = 16777216$
Number of cells = $2^L =$ 16
Expected number per cell = 1048576
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

$\Delta = 1$, $\mu = 2.0861626e-07$, $\sigma = 1$

Test Results for $\Delta = 1.0000$

Number of degrees of freedom	:	8
Value of the statistic	:	9.00
p-value of test	:	0.34

CPU time used : 00:00:00.00

Generator state:
486539008 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 110.21
p-value of test : 0.87

CPU time used : 00:00:00.01

Generator state:

503316224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32543.34
p-value of test : 0.81

CPU time used : 00:00:00.02

Generator state:
520093440 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 12887.84
p-value of test : eps *****

CPU time used : 00:00:00.00

Generator state:
536870656 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 5225.13
p-value of test : eps *****

CPU time used : 00:00:00.01

Generator state:

553647872 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 1.17

p-value of test : 0.12

CPU time used : 00:00:00.01

Generator state:

570425088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32

ChiSquare statistic : 3099.74

p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 31

ChiSquare statistic : 1379.35

p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom	:	32	
ChiSquare statistic	:	370.72	
p-value of test	:	eps	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	25	
ChiSquare statistic	:	37.84	
p-value of test	:	0.05	

Test on the values of the Statistic C

Number of degrees of freedom	:	15	
ChiSquare statistic	:	27.82	
p-value of test	:	0.02	

CPU time used : 00:00:00.07

Generator state:
587202304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	1343.36	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	634.67	
p-value of test	:	eps	*****

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	0.50
p-value of test	:	0.78

CPU time used : 00:00:00.01

Generator state:

620756480 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 8.95
p-value of test : 0.35

CPU time used : 00:00:00.01

Generator state:
637533696 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 131.91
p-value of test : 0.39

CPU time used : 00:00:00.00

Generator state:
654310912 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

```

-----
N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
Delta = 1, Mu = 0.00097653271, Sigma = 1

```

```

-----
Test Results for Delta = 1.0000

```

```

Number of degrees of freedom : 32768
Value of the statistic : 32492.05
p-value of test : 0.86

```

```

-----
CPU time used : 00:00:00.02

```

```

Generator state:
671088128 bits have been read.

```

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

```

```

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

```

```

sstring_HammingIndep test:

```

```

-----
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

```

```

Counters with expected numbers >= 10

```

```

-----
Number of degrees of freedom : 177
Chi-square statistic : 167.29
p-value of test : 0.69

```

```

-----
CPU time used : 00:00:00.01

```

```

Generator state:
687865344 bits have been read.

```

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

```

```

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

```

unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 282.66
p-value of test : 0.82

CPU time used : 00:00:00.00

Generator state:

704642560 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.54
p-value of test : 0.71

CPU time used : 00:00:00.01

Generator state:

721419776 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32	
ChiSquare statistic	:	71.28	
p-value of test	:	8.1e-5	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	31	
ChiSquare statistic	:	37.71	
p-value of test	:	0.19	

Test on the values of the Statistic J

Number of degrees of freedom	:	32	
ChiSquare statistic	:	31.37	
p-value of test	:	0.50	

Test on the values of the Statistic R

Number of degrees of freedom	:	25	
ChiSquare statistic	:	29.79	
p-value of test	:	0.23	

Test on the values of the Statistic C

Number of degrees of freedom	:	15	
ChiSquare statistic	:	19.65	
p-value of test	:	0.19	

CPU time used : 00:00:00.07

Generator state:
738196992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom : 62
ChiSquare statistic : 1818.36
p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 62
ChiSquare statistic : 940.51
p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom : 160
ChiSquare statistic : 288.79
p-value of test : 1.9e-9 *****

Test on the values of the Statistic R

Number of degrees of freedom : 57
ChiSquare statistic : 60.61
p-value of test : 0.35

Test on the values of the Statistic C

Number of degrees of freedom : 32
ChiSquare statistic : 31.20
p-value of test : 0.51

CPU time used : 00:00:00.05

Generator state:
754973952 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
8 RandomWalk1 H (L = 64)	8.1e-5

```
9 RandomWalk1 H (L = 320)      eps
9 RandomWalk1 M (L = 320)      eps
9 RandomWalk1 J (L = 320)      1.9e-9
```

All other tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alhabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 0.087
p-value of test : 0.96

CPU time used : 00:00:00.01

Generator state:

771751168 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,

Sparse = FALSE

Number of bits = n = 16777216

Number of cells = 2^L = 16

Expected number per cell = 1048576

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	8
Value of the statistic	:	7.07
p-value of test	:	0.53

CPU time used : 00:00:00.00

Generator state:

788528384 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216

Number of cells = 2^L = 256

Expected number per cell = 65536

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	128
Value of the statistic	:	121.02
p-value of test	:	0.66

CPU time used : 00:00:00.01

Generator state:

805305600 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32891.63
p-value of test : 0.31

CPU time used : 00:00:00.02

Generator state:
822082816 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 188.00
p-value of test : 0.27

CPU time used : 00:00:00.01

Generator state:
838860032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	337.75
p-value of test	:	0.10

CPU time used	:	00:00:00.00
---------------	---	-------------

Generator state:
855637248 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	1.92
p-value of test	:	0.03

CPU time used	:	00:00:00.01
---------------	---	-------------

Generator state:
872414464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	28.12
p-value of test	:	0.66

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	18.21
p-value of test	:	0.97

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	41.99
p-value of test	:	0.11

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	28.97
p-value of test	:	0.27

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	13.23
p-value of test	:	0.58

CPU time used : 00:00:00.06

Generator state:
889191680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32
```

swalk_RandomWalk1 test:

```
-----
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320
```

Test on the values of the Statistic H

```
Number of degrees of freedom      : 62
ChiSquare statistic                : 1933.79
p-value of test                   : eps      *****
```

Test on the values of the Statistic M

```
Number of degrees of freedom      : 62
ChiSquare statistic                : 919.00
p-value of test                   : eps      *****
```

Test on the values of the Statistic J

```
Number of degrees of freedom      : 160
ChiSquare statistic                : 281.50
p-value of test                   : 1.0e-8   *****
```

Test on the values of the Statistic R

```
Number of degrees of freedom      : 57
ChiSquare statistic                : 69.80
p-value of test                   : 0.12
```

Test on the values of the Statistic C

```
Number of degrees of freedom      : 32
ChiSquare statistic                : 37.47
p-value of test                   : 0.23
```

CPU time used : 00:00:00.06

Generator state:

905968640 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	1.0e-8
-----	-----

All other tests were passed